

Quiz 3-1 [25 min]

Check **ONLY T** if the given statement in the question is correct or **ONLY F** if not.

Correct Answer 4pts, No Answer -1 pts, Wrong Answer -2 pts. If the total points < 0, your score will be set to 0.

1. To show that if S is a finite set with n elements where n is a nonnegative integer, then S has 2^n subsets, let T be a set with $k + 1$ elements. Then, we can write $T = S \cup \{a\}$ where $a \in T$ and $S = T - \{a\}$. For each subset X of S there are exactly two subsets of T . Because there are 2^k subsets of S , there are $2 \cdot 2^k = 2^{k+1}$ subsets of T . **(F)**

2. For integer n , $1^3 + 2^3 + \dots + n^3$ is $O(n^4)$. **(T)**

3. Suppose that $f(x)$ is $O(g(x))$. It follows that $2^{f(x)}$ is $O(2^{g(x)})$. **(F)**

4. If A , B , and C are sets, then $A - (B \cap C) = (A - B) \cap (A - C)$. **(F) / $A = \{1,2\}$, $B = \{1\}$, $C = \{2\}$**

5. $F : \mathbb{Z} \rightarrow \mathbb{Z}$ where $F(x) = \frac{1}{x^2 - 5}$. **(F) / F(1) is not integer.**

6. The f is a function from the set of all bit strings to the set of integers if $f(S)$ is the position of a 1 bit in the string S .

(F) / There may be no 1 bits or more than one 1 bit.

7. The following argument is valid. **(T)**

- Rainy days make gardens grow.
- Gardens don't grow if it is not hot.
- It always rains on a day that is not hot.
- Therefore, if it is not hot, then it is hot.

8. In a group of ten people (where any two people are either friends or enemies) there are either three mutual friends or four mutual enemies. Let A be one of the people. A either has at least four friends or else has at least six enemies among the other nine people.

Case 1: A has at least four friends, say B, C, D, E . If any two of B, C, D, E are friends, then these two together with A form a group of three mutual friends. If none of B, C, D, E are friends with each other, then B, C, D, E are four mutual enemies.

Case 2: A has at least six enemies, say B, C, D, E, F, G . Applying the Pigeonhole Principle to this set of six, there are either three mutual friends or three mutual enemies. If there are three friends, we are done. If there are three mutual enemies, then these three together with A form a group of four mutual enemies.

(T)

9. For k greater than 2, a function f from a set with $k + 1$ or more elements to a set with k elements is not one-to-one. **(T)**

10. x is positive integer. The function $f(x) = (x + 2) \log(x^2 + 1) + \log(x^3 + 1)$ is $O(x \log x)$. **(T)**

11. The following proof for the statement, “For every nonnegative n , $5n = 0$ ”, by strong induction is Correct. **(F)**

- Basis step : $5 \cdot 0 = 0$.

- Inductive step : Suppose that $5j = 0$ for all nonnegative integers j with $0 \leq j \leq k$. Write $k + 1 = i + j$, where i and j are natural numbers less than $k + 1$. By the inductive hypothesis, $5(k + 1) = 5(i + j) = 5i + 5j = 0 + 0 = 0$.

12. $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$, $A \odot B = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$. **(T)**